

Coding & Solution

Date	15 March 2026
Team ID	xxxxxx
Project Name	Credit Card Auto-Approval Prediction System
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	github.com/RevanthBoina/Credit-card-Auto-Approval-prediction
Programming Language(s)	[Python, TypeScript, HTML5, CSS3 , correctly specifies Python and Flask for the backend (which runs the ML model) and Next.js/TypeScript for the frontend files.
Framework(s) Used	Flask (Python backend engine), Next.js (Frontend interface framework)
Key Features Implemented	• Serialized machine learning pipeline (Random Forest Model, 95.8% accuracy). • User-friendly input form collecting demographic and financial parameters. • Real-time prediction display showing application approval or rejection results alongside model confidence probability metrics. • Automated data visualization charts (Exploratory Data Analysis - EDA) covering age, gender, education, and income distribution features.
Pending / Incomplete Features	None. All defined core functional requirements for the data pipeline and interactive prediction dashboard have been completed and verified.
Setup / Run Instructions	Local Python Flask Server Setup: 1. Navigate to directory: cd web 2. Install dependencies: pip install -r requirements.txt 3. Launch web server application: python app.py 4. Open browser to: http://127.0.0.1:8080

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The system successfully implements the complete lifecycle of a classification task:

1. Data ingestion and preprocessing (handling null structures and categorical transforms).
 2. Model training and evaluation (comparing Logistic Regression, Decision Trees, and Random Forests).
 3. Serialization of the best-performing pipeline using Joblib.
 4. Serving predictions dynamically using Flask, rendering results cleanly via semantic templates.
- The codebase is clean, fully modularized, and follows PEP 8 standards.